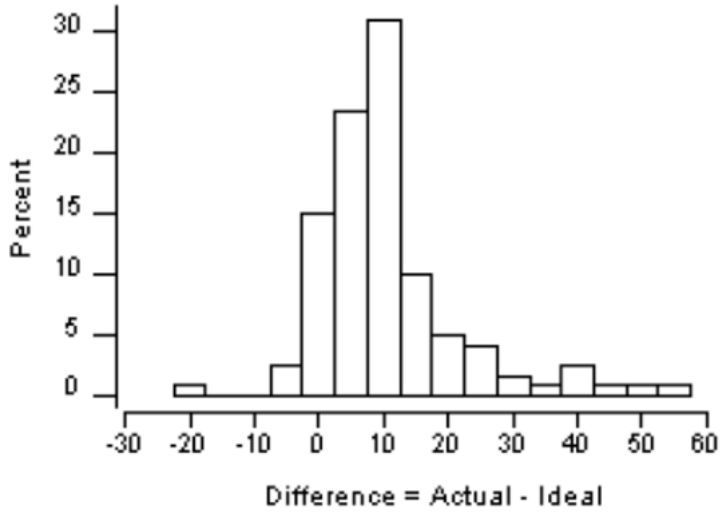
Circle the correct answer for each question. If you're unsure, you can write numbers 1, 2, 3, etc. next to the choices to get partial credit.

The following histogram shows the distribution of the difference between the actual and "ideal" weights for 119 female students. Ideal weights are responses to the question "What is your ideal weight"? The difference = actual – ideal.



1. What is the approximate shape of the distribution?

   (a) The median and mean are approximately equal.
   (b) The median is larger than the mean.
   (c) The mean is larger than the median.
   (d) The mean and the median cannot be compared without calculations.

   Correct answers: (c)

2. How does the mean compare to the median in this data set?

   (a) The median is larger than the mean.
   (b) The mean and the median cannot be compared without calculations.
   (c) The mean is larger than the median.
   (d) The median and mean are approximately equal.

   Correct answers: (c)

3. If the observation at -20 pounds was removed from the data set, the mean _____.

   (a) may or may not change
   (b) would remain the same
   (c) would decrease
   (d) would increase

   Correct answers: (d)

A hot dog manufacturer asserts that a brand of hot dogs has an average fat content of 18 grams per hot dog. Fat contents in hot dogs from this manufacturer follow an approximately normal distribution with a standard deviation of 1.1 grams. Define the statistic x = mean fat content in a random sample of 10 hot dogs.

4. The value 18 is the _____.

   (a) parameter
   (b) statistic
   (c) standard deviation
   (d) sample size

   Correct answers: (b)

5. Is the sampling distribution of x approximately normal?

   (a) No, since the sample size is too small for the Central Limit Theorem to hold.
   (b) Yes, since the population of fat contents in hot dogs is approximately normal.
   (c) Not enough information exists to determine if the sampling distribution of x is approximately normal.
   (d) No, since the manufacturer does not know if the population of hot dogs is more than 20 times the size of the sample.

   Correct answers: (b)

6. Which of the following distributions would have the smallest standard deviation?

   (a) Number of miles put on a rental car in one day
   (b) Average number of miles put on a rental car per day across 5 customers
   (c) Average number of miles put on a rental car per day across 25 customers
   (d) Distributions of all of the above would have the same standard deviation.

   Correct answers: (c)